

A living planet

The existence of complex life is what makes Earth unique in the Solar System. A combination of lucky accidents gave the planet the right conditions for life, and a series of chemical reactions created the first microscopic living things. From a slow start, life evolved into a dazzling diversity of forms that live all over the globe.

WATER OF LIFE

Liquid water is essential for life. Luckily for us, Earth's distance from the Sun, along with the insulating effect of its atmosphere, gives it an average temperature of about 59°F (15°C). This allows liquid water to flow almost everywhere on the planet. By contrast, all the other planets in the Solar System are either much too hot or much too cold.

▼ VITAL SUPPLY

Liquid water is the one thing all life forms need, from the smallest microbe to the tallest trees of this tropical forest.



CHEMISTRY OF LIFE

Water is vital because it mixes with so many other substances to make the right chemistry for life. This chemistry takes place inside cells—microscopic bags of fluid that turn chemicals into complex molecules such as proteins.



▲ LIVING CELL

This Paramecium is a microscopic single-celled organism. Plants and animals are made up of many cells.



ORIGINS

Life began about 3.8 billion years ago, possibly when a chance chemical reaction created special molecules called amino acids. This may have been triggered by a lightning strike. Another theory is that these molecules fell to Earth from space. Amino acids link together to form proteins—the basis of all living things.